**Example 1.** Prove that, if $n$ is greater than $2$, then $\phi(n)$ is even.

**Solution.** We split into cases.

**Case 1.** $n$ is a power–of–$2$ greater than $2$. Then

$n =$

**Case 2.** $n$ has an prime factor $p$ that is:

$n =$

**Example 2.** Find all positive integers $n$ with $\phi(n) = 12$.

**Solution.** As $n$ need be greater than $1$, take prime decomposition

$n = p_1^{k_1} p_2^{k_2} \cdots p_r^{k_r}$

Then:

$12 = \phi(n) =$

This leads us to some simple observations.

which cuts the form of $n$ down to:

$n =$

Addressing all integers of this form, we solve $\phi(n) = 12$ to get

largest prime factor is $13 \rightarrow$

largest prime factor is $7 \rightarrow$

largest prime factor is $5 \rightarrow$

largest prime factor is $3 \rightarrow$

largest prime factor is $2 \rightarrow$

**Example 3.** Let $n$ be an odd integer. Prove that $\phi(n) \geqslant \sqrt{n}$.

**Proof.** Both sides are multiplicative functions of $n$. The inequality is true if $n = 1$.

Since, further, the values of these functions are nonnegative, and hence preserve directions of inequalities under multiplication, it suffices to prove

$\rightarrow$ **goal:**

Let $p$ be an odd prime. We use induction on $k$.

**Base case.** If $k = 1$, we need to show

But this results from the following lemma, as $p \geqslant 3$ as an odd prime.

**Lemma.** $n - 1 \geqslant \sqrt{n}$ for integers $n$ greater or equal $3$.

**Proof.** Rewrite the inequality as the equivalent

Evidently the inequality holds if $n = 3$.

Further, the lefthand side:

for example as we can see by factoring it as:

Therefore $n - \sqrt{n}$ starts at $n = 3$ with value $\geqslant 1$, and increases with $n$, so

□

**Inductive Assumption.** Let pos. integer $k$ be given, and

assume $\rightarrow$

**Inductive Step.** Then

$\phi(p^{k+1}) =$

This completes the inductive step, and hence the proof. □

A. **Euler's Theorem.** We generalize Fermat's Theorem that $a^{p-1} \equiv 1 \mod p$.

**Euler's Theorem.** Let $n$ be a positive integer and $a \perp n$ be an integer. Then

**Proof.** First let's reduce our work to the case that $n$ is a power of a prime.

**Lemma.** If the theorem holds for moduli $m \perp n$, then it also hold for modulus:

**Proof.** Assume the theorem holds for moduli $m \perp n$.

Let $a \perp mn$ be given. Then:

Next, since we have assumed the theorem holds for moduli $m$ and $n$ we have

$a^{\phi(m)} \equiv$

$a^{\phi(n)} \equiv$

Next, because $m \perp n$, this system implies

□

Using this lemma, and the fact that the theorem obviously holds for $n = 1$, it

Let modulus $p^k$ be a positive power of a prime and $a \perp p$. We induct on $k$.

**Base Case.** The case $k = 1$ is:

**Inductive Assumption.** Let integer $k$ be pos. and

assume $\rightarrow$

**Inductive Step.** We must show

goal $\rightarrow$

Note that: $\phi(p^{k+1}) =$

so equivalently we must show

goal $\rightarrow$

This is an immediate consequence of the following lemma, using $b =$

**Lemma.** If $b \equiv 1 \mod p^k$, then $b^p \equiv 1 \mod p^{k+1}$.

We prove the lemma on the following page.

**Lemma.** If $b \equiv 1 \mod p^k$, then $b^p \equiv 1 \mod p^{k+1}$.

**Proof.** We factor using the geometric sum formula

$b^p - 1 =$

Because $b \equiv 1 \mod p^k$ it follows that the first factor $(b-1)$ is

On the other hand, $b \equiv 1 \mod p^k$ also implies:

and thus

$1 + b + b^2 + \cdots + b^{p-1} \equiv$

Hence the second factor $(1 + b + b^2 + \cdots + b^{p-1})$ is

Thus the product of the factors, $b^p - 1$, is divisible by

And hence $b^p \equiv 1 \mod p^{k+1}$.

□